

현대자동차 AI 과제 - 비전팀

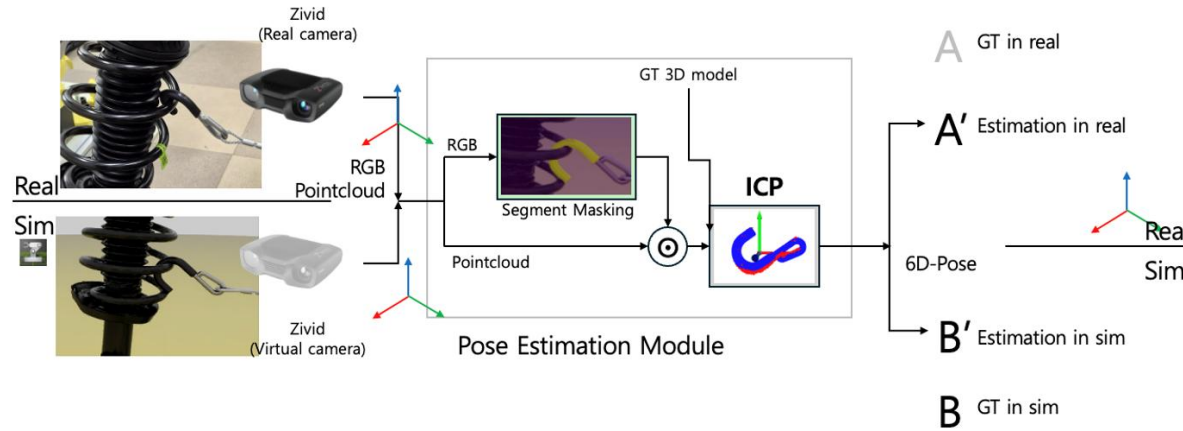
2026.04.29.

Presenter : Hyun Woong Choi

Vision팀 목표

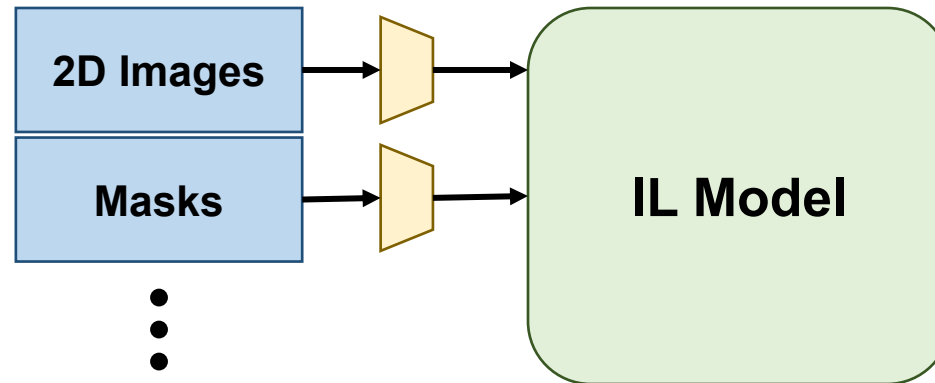
▪ 1) Hook 6D Pose Estimation

- Real 환경에서 Hook의 정확한 6D Pose 추출 및 Sim을 이용한 검증



▪ 2) Images, Segmentation Masks 데이터수집 for IL

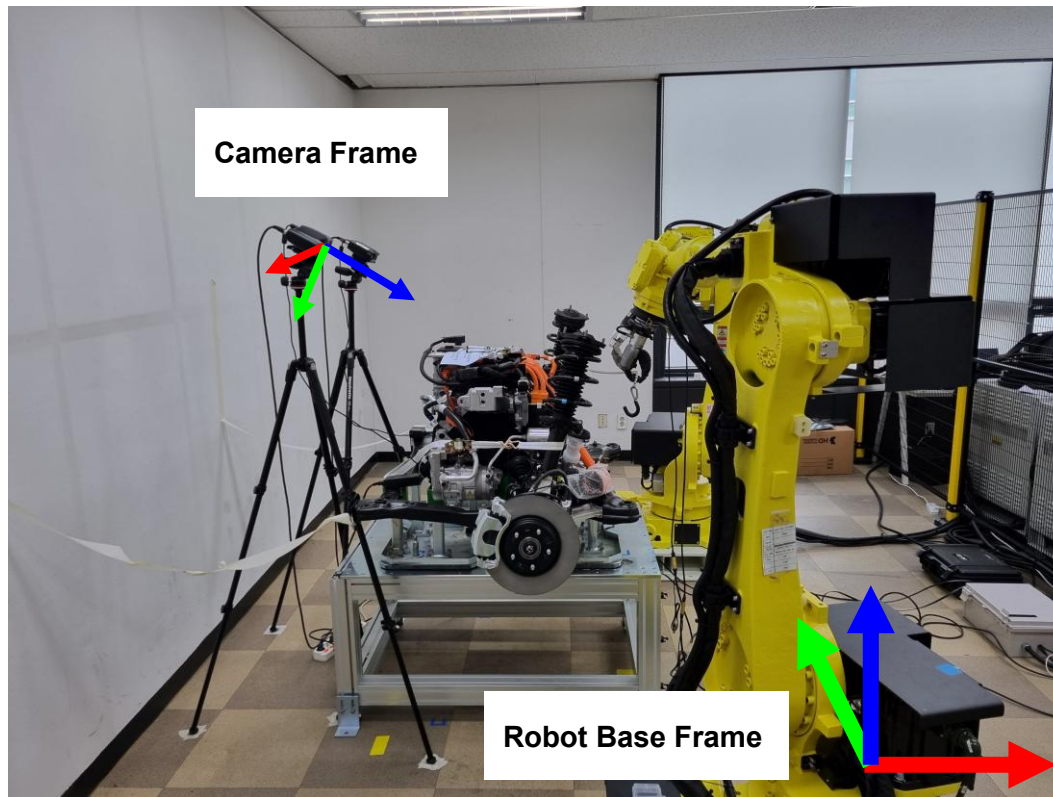
- Real 환경의 Zivid, Wrist Cam 카메라 영상 및 Hook 마스크 수집



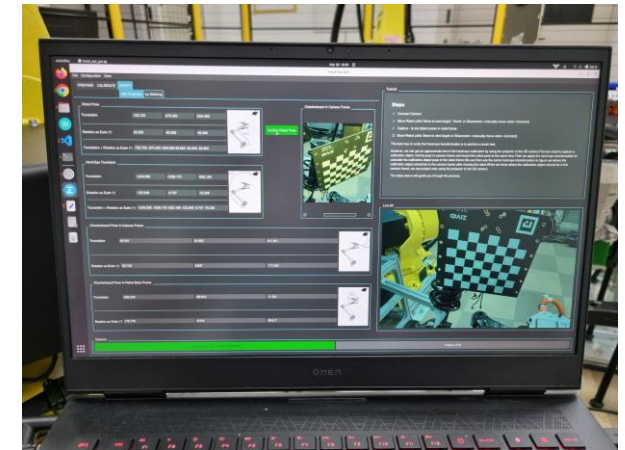
Vision 카메라 세팅

▪ Zivid Camera Setting & Calibration

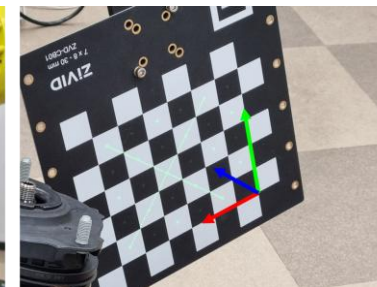
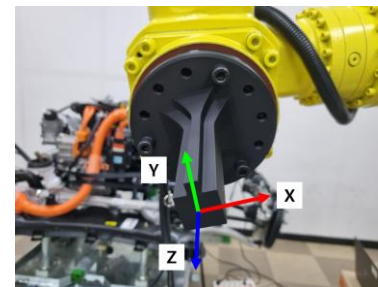
- 최적의 위치에 카메라 설치
- 체커보드를 이용한 Calibration 및 Verification 테스트 진행



$$p^{Base} = T_{Camera}^{Base} \cdot p^{Camera}$$



Checkerboard Calibration

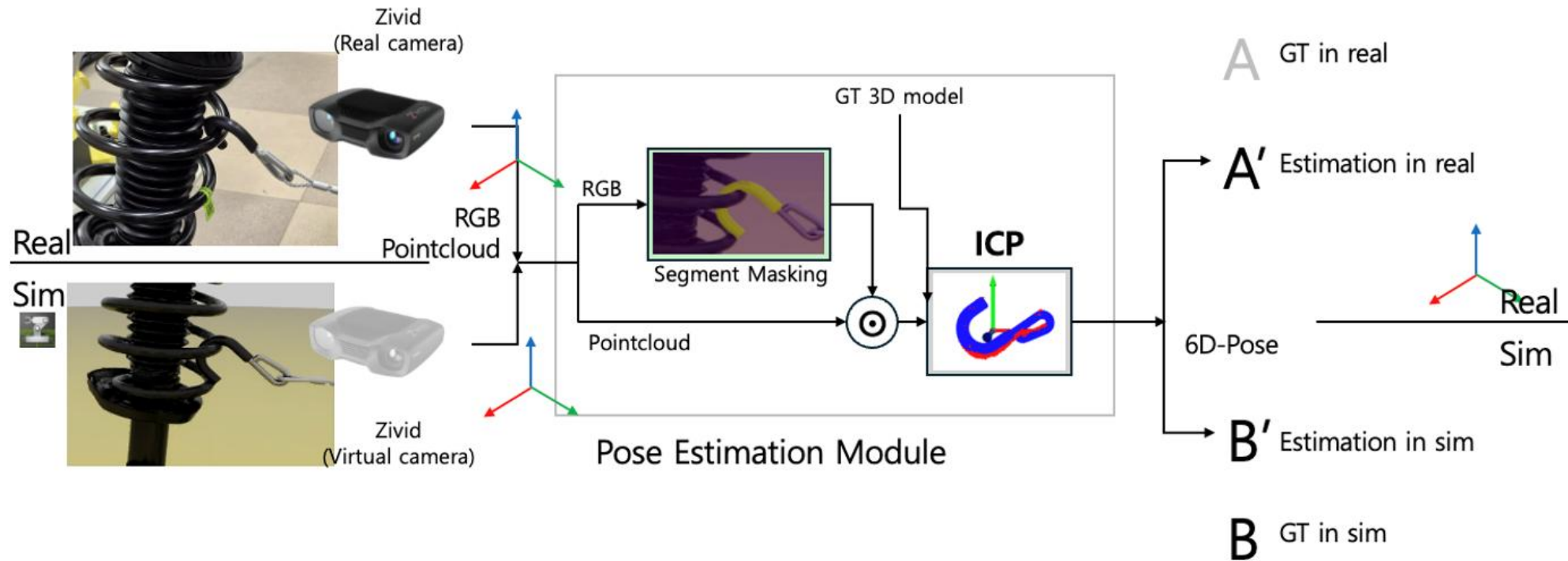


Calibration Verification

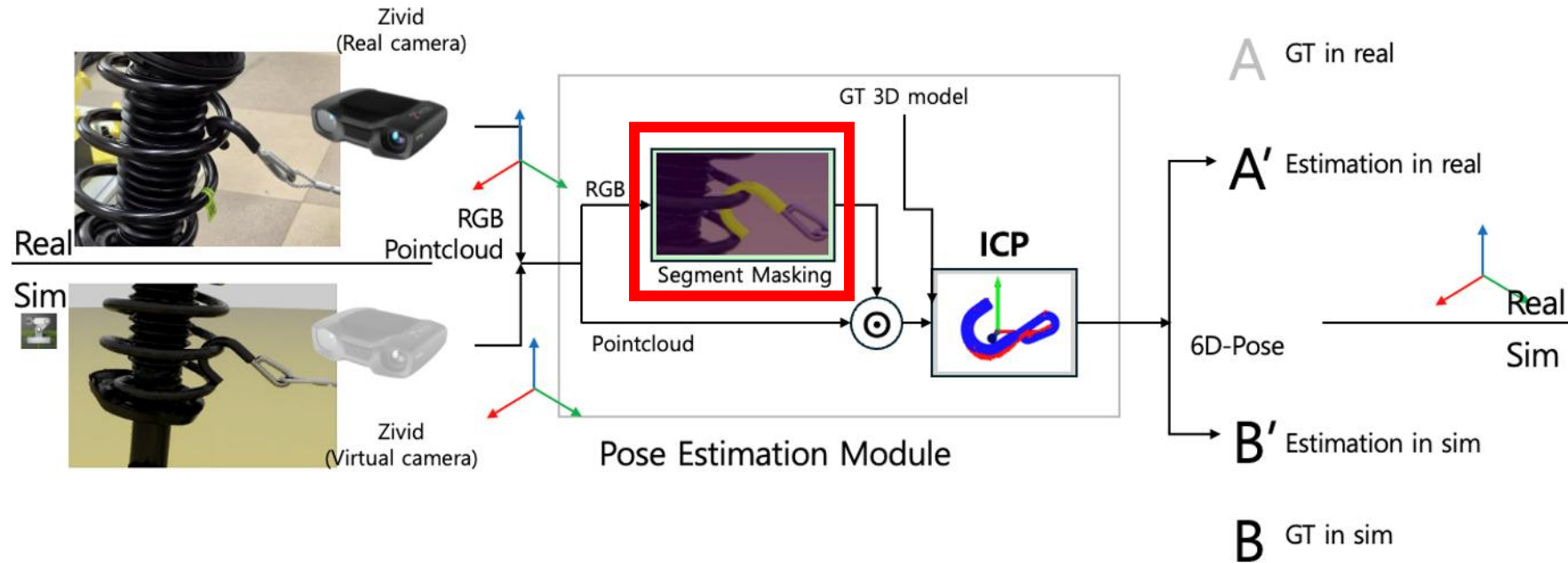
1) Hook 6D Pose Estimation

▪ Hook Pose Estimation Pipeline

- Step1) Zivid 카메라를 통해 RGB 이미지 및 3D Pointcloud 동시 획득
- Step2) RGB 이미지를 AI 모델에 입력하여 Hook 영역의 정밀 Mask 추출
- Step3) 추출된 Mask에 해당하는 PointCloud만 분리한 후, 사전 제작된 Hook 3D Model과 ICP(Iterative Closest Point) 정합 수행



1) Hook Segmentation



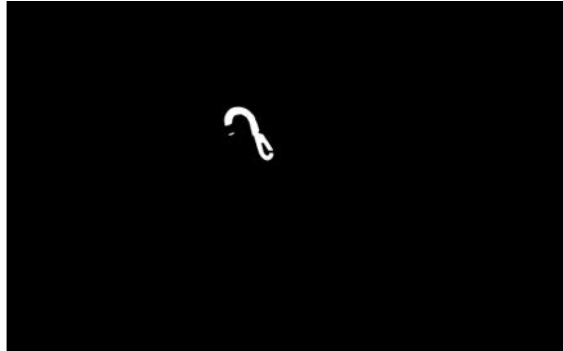
1) Hook Segmentation

▪ 다양한 후크 데이터셋 확보

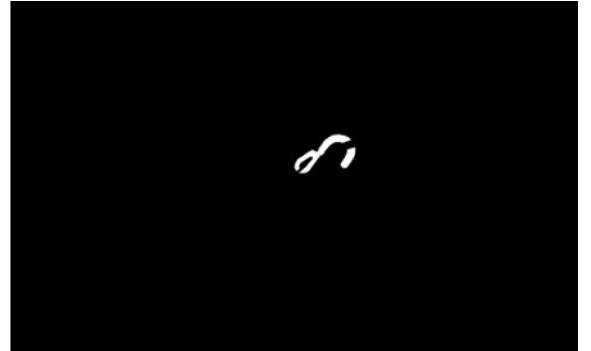
- 총 데이터셋 880장 확보 후 Labeling



Left Camera



Right Camera



- 환경 외란에 강건하게 하기 위해 조도, 배경, Occlusion 등 다양한요소 반영



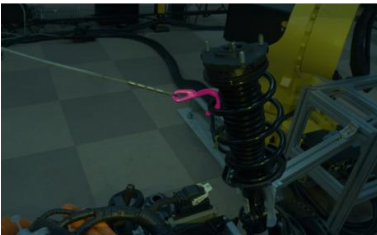
1) Hook Segmentation

Segmentation AI 모델 튜닝

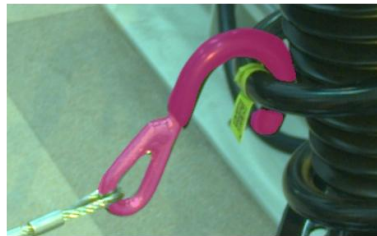
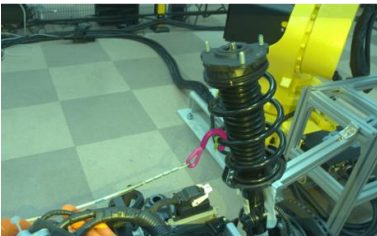
- 6개의 Segmentation 모델 튜닝 (YOLO, SAM3, RF-DETR, Mask R-CNN, Mask2Former, PointRend)
- 테스트 결과 성능이 가장 준수하게 나온 RF-DETR 모델 선택

Images

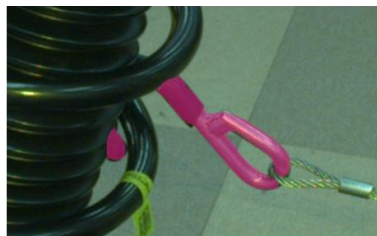
Test Results



Test Image #16

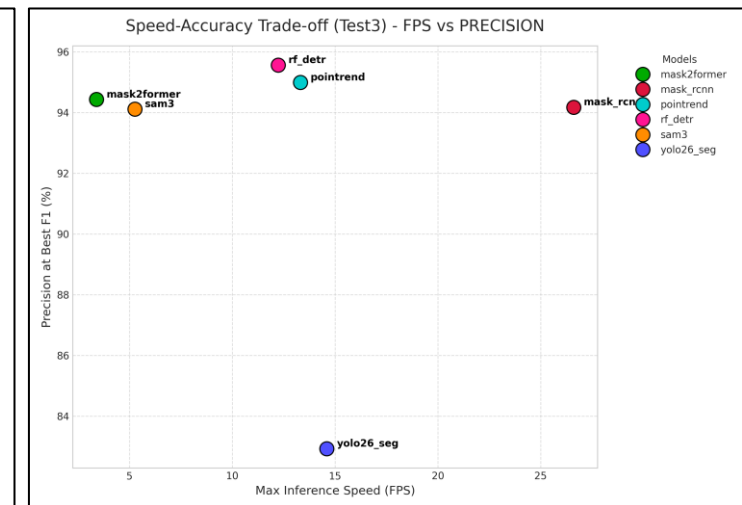
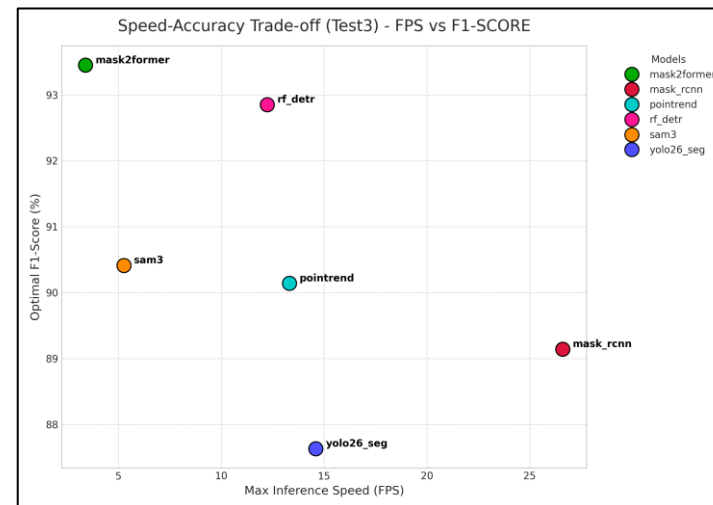


Test Image #24

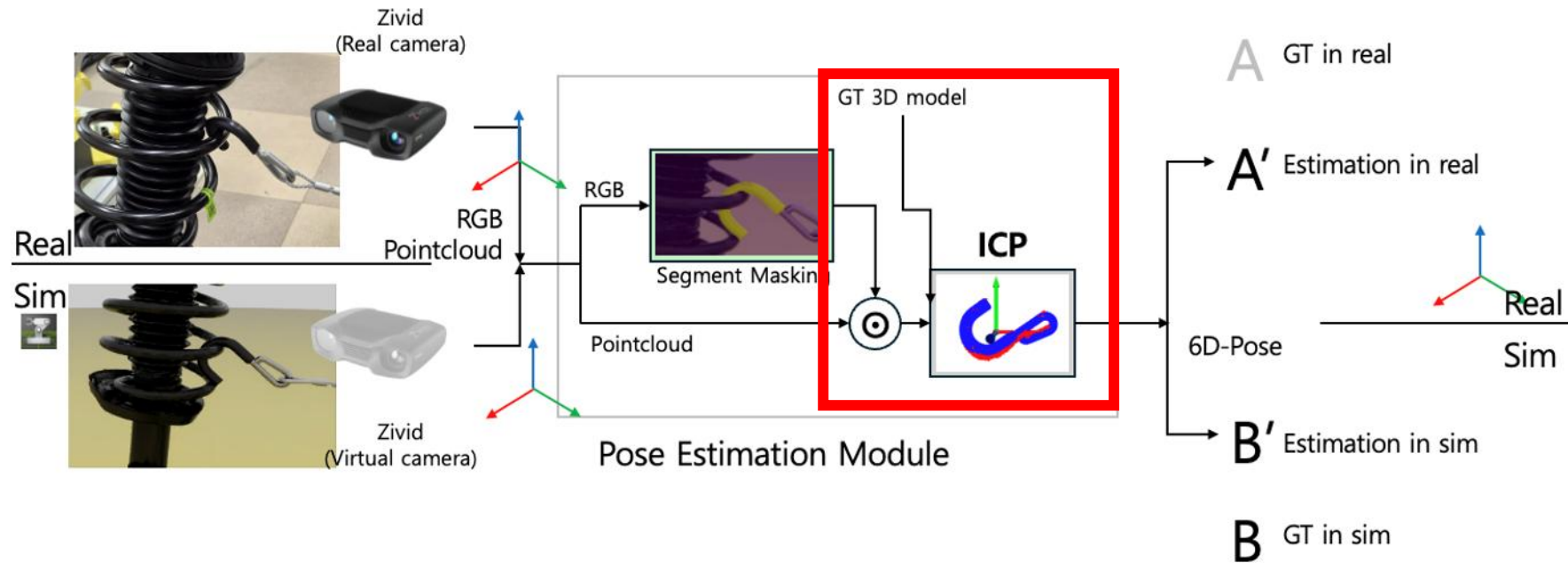


Test Image #37

Models	Precision	Recall	Detection Rate (%)	F1-Score (%)	Boundary IoU (%)	HD95 (px)	Estimated FPS
Mask R-CNN	94.17	85.28	100	89.14	25.59	13.15	11.02
Mask2Former	94.43	92.98	100	93.45	39.55	8.63	2.94
PointRend	94.99	86.88	100	90.14	29.98	9.24	10.93
RF-DETR	95.56	90.70	100	92.85	35.82	8.05	8.94
SAM3	94.11	87.92	100	90.41	29.22	13.05	4.39
Yolo26-seg	82.92	94.30	100	87.63	18.95	12.01	13.51



1) ICP 기반 6D Pose Estimation



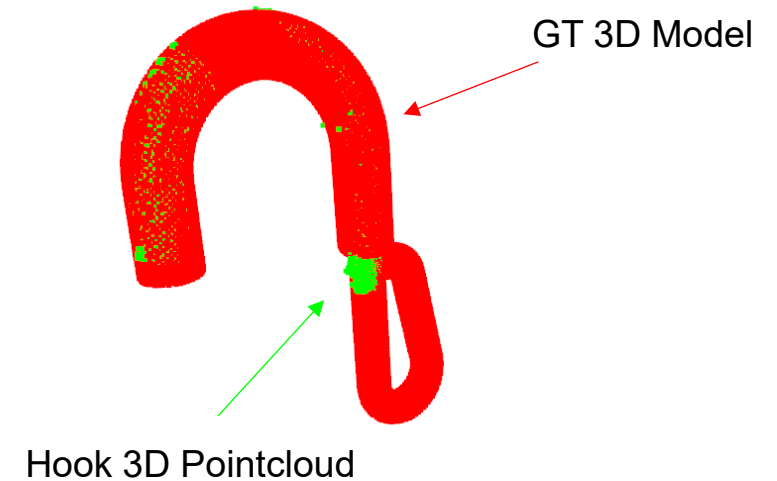
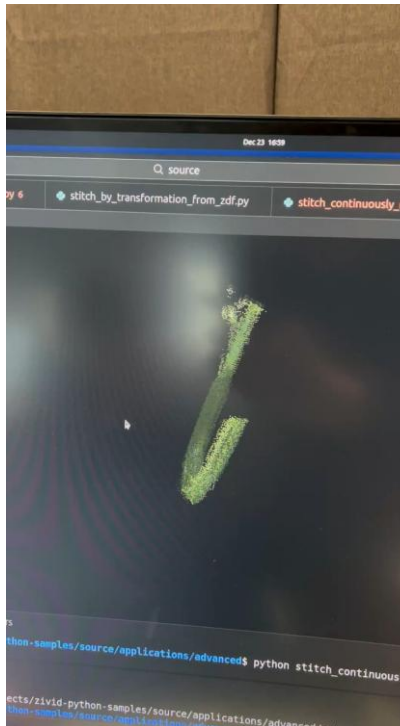
1) ICP 기반 6D Pose Estimation

▪ Hook 3D Modeling (Ground Truth)

- Zivid 카메라를 활용하여 실물 Hook 정밀 3D 스캐닝
- Hook를 회전시키며 100 프레임의 Pointcloud 획득 후, 병합을 통해 3D Model 구축

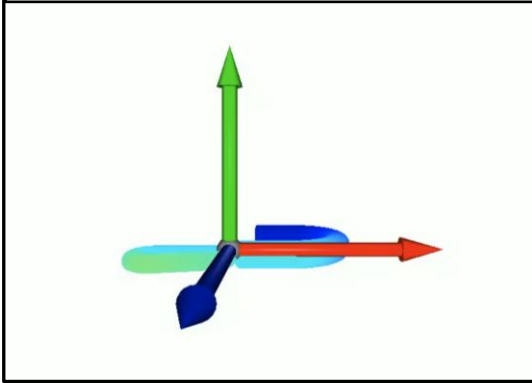
▪ ICP (Iterative Closest Point) 알고리즘 적용

- 사전 구축된 3D Reference 모델과 현재 Zivid로 획득한 부분 Pointcloud 간의 최적 정합점 도출 및 최종 자세 산출



1) Hook 6D Pose Estimation Results

1.Hook 3d model



2.RGB



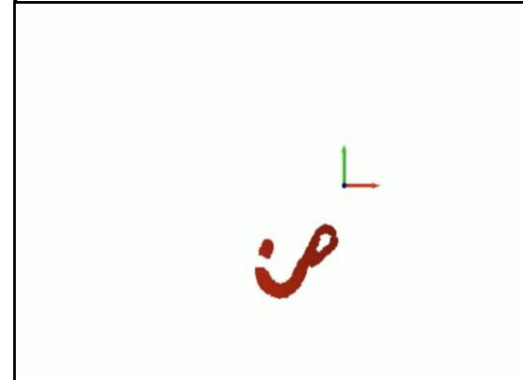
3.Hook mask



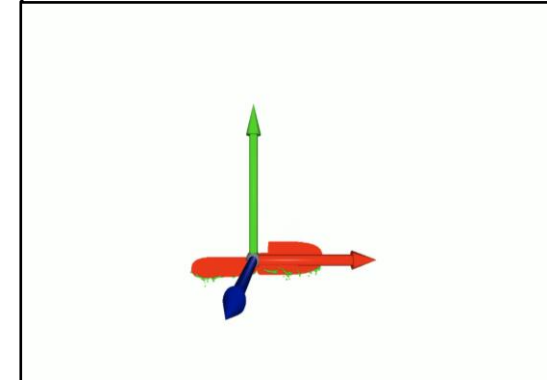
4.Mask overlayed RGB



5.Masked pointcloud



6.ICP result



1) Hook 6D Pose Estimation Results

7. Pose estimation result



2) Dataset for IL

▪ Zivid & Wrist Cam Video Record

- Real 환경 데이터 수집 시 Zivid, Wrist cam 영상 동시 녹화 기능 추가



Zivid



Realsense

Future Work

▪ Hook 6d Pose Estimation

- 다양한 환경 (조도 변화, 가림 등)에서의 기존 파이프라인 강건성 테스트
- Sim 환경 내의 Ground Truth와의 오차 비교를 통한 정량적 성능 검증
- 추론 속도 및 정확도 향상을 위해 End-to-End Model(Foundation Pose 등)의 커스텀 데이터로 파인튜닝 시도

▪ Images, Masks 획득 for IL

- Real에서 수집한 영상 대상 Hook Segmentation Mask 후처리 작업
- 로봇 핸드에 의한 Occlusion현상에 강건한 Segmentation Model 정밀 튜닝

